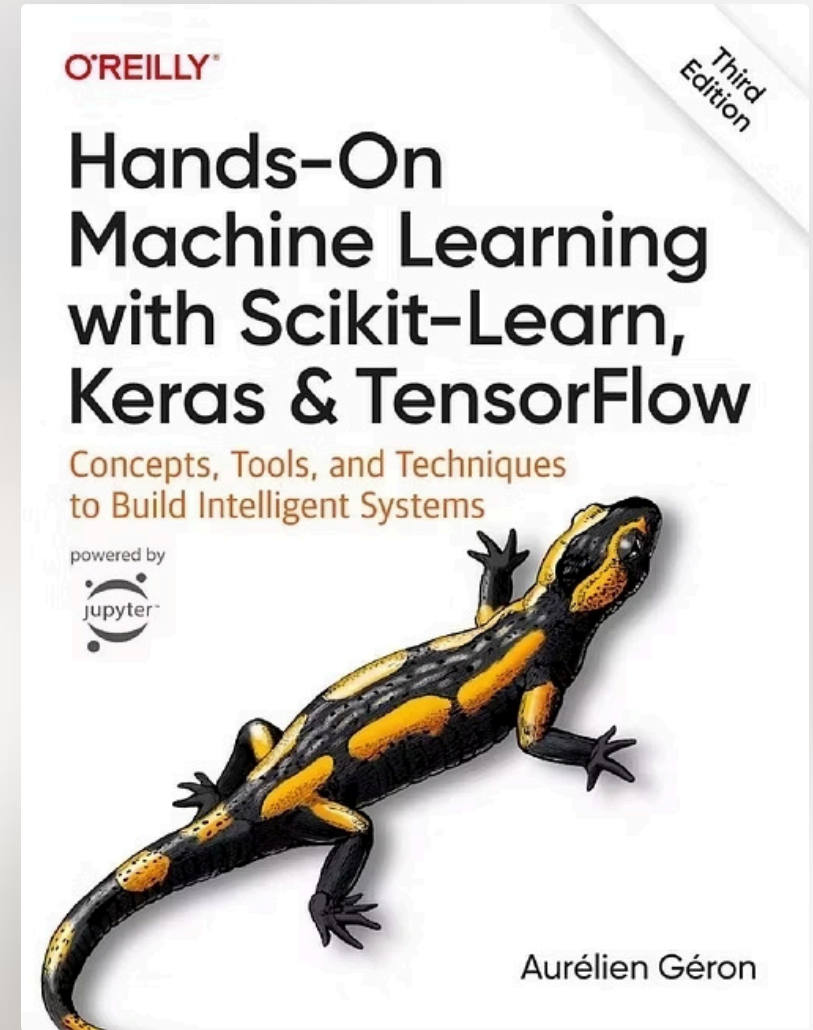


# Machine Learning

Lessons based on the Textbook "Hands-On Machine Learning with Scikit-Learn, Keras, and TensorFlow" by Aurélien Géron, 2023.

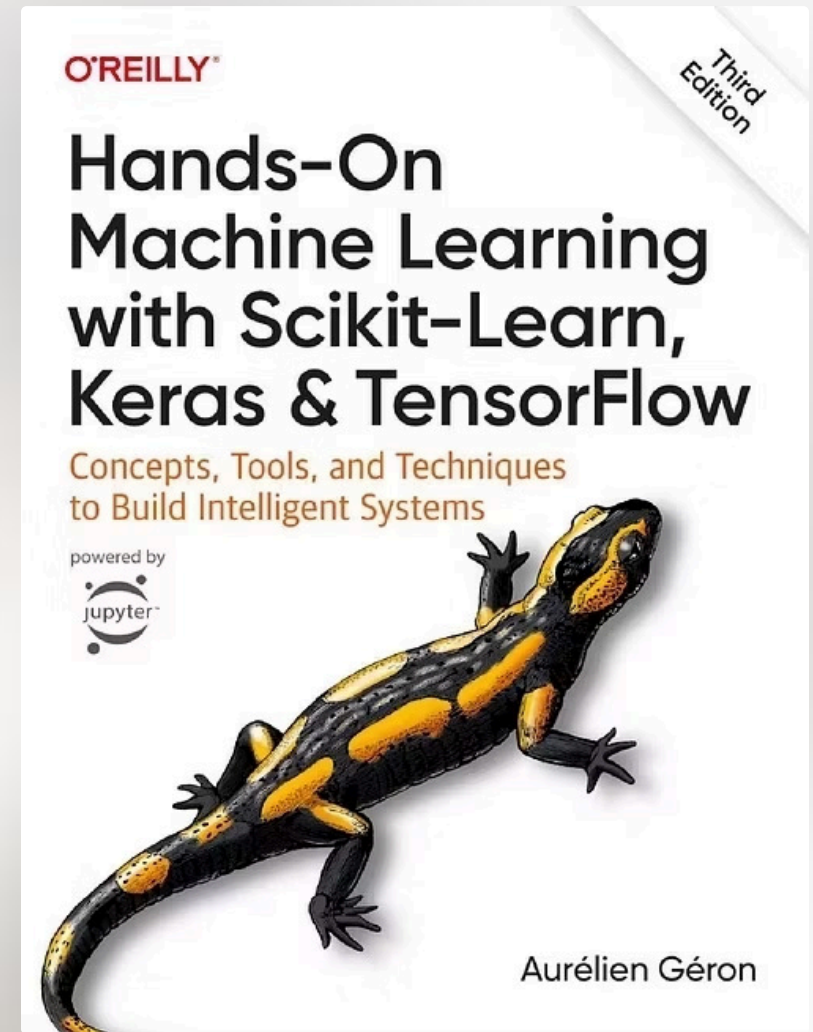
**RP** by Rodrigo Martins Pagliares

Last update: 07/01/2025



# Machine Learning

## Chapter 1 – The Machine Learning Landscape





# The Machine Learning Landscape

**ML** has moved beyond **science fiction** to become integral to **everyday technology**.

Early applications, like **Optical Character Recognition (OCR)**, were specialized.

The **first mainstream ML** application: **Spam Filters** in the **1990s**, which improved **email management**.

**ML** now powers a wide range of services like **voice assistants**, **automatic translations**, and **recommendation systems**.

# Agenda for Chapter 1

1. What Is Machine Learning?
2. Why Use Machine Learning?
3. Examples of Applications
4. Types of Machine Learning Systems
5. Main Challenges of Machine Learning
6. Testing and Validating





# Agenda for Chapter 1

## 1. What Is Machine Learning?

## 2. Why Use Machine Learning?

## 3. Examples of Applications

## 4. Types of Machine Learning Systems

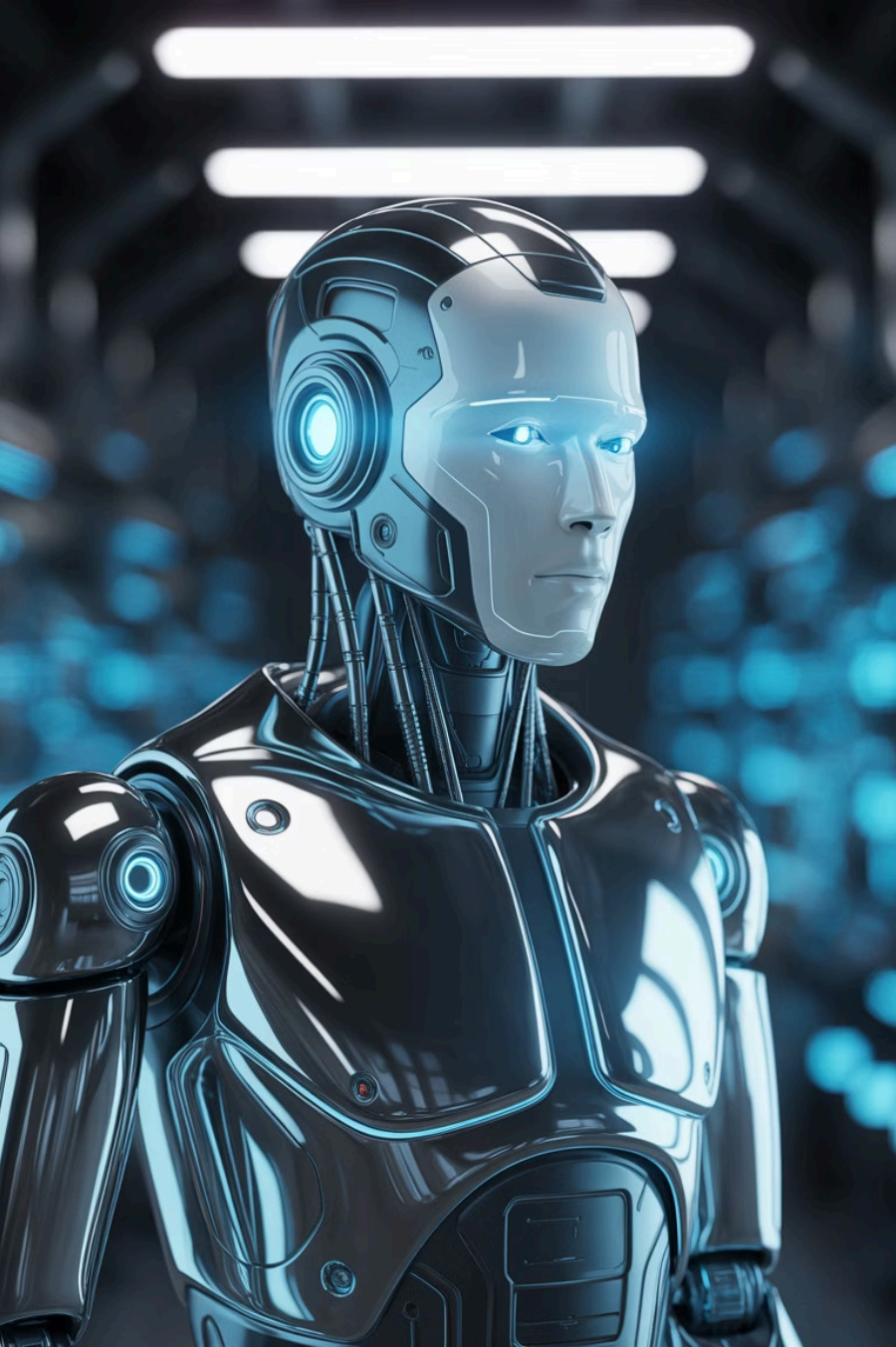
- Classification criteria #1: Supervision during training
  - Supervised, Unsupervised, Semi-Supervised, Self-Supervised, and Reinforcement Learning
- Classification criteria #2: Approach to Learning Method
  - Batch and Online (Incremental) Learning
- Classification criteria #3: Learning Method
  - Instance-Based and Model-Based Learning

## 5. Main Challenges of Machine Learning

- Insufficient Quantity of Training Data
- Non-representative Training Data
- Poor-Quality Data
- Irrelevant Features
- Overfitting the Training Data
- Underfitting the Training Data
- Stepping Back

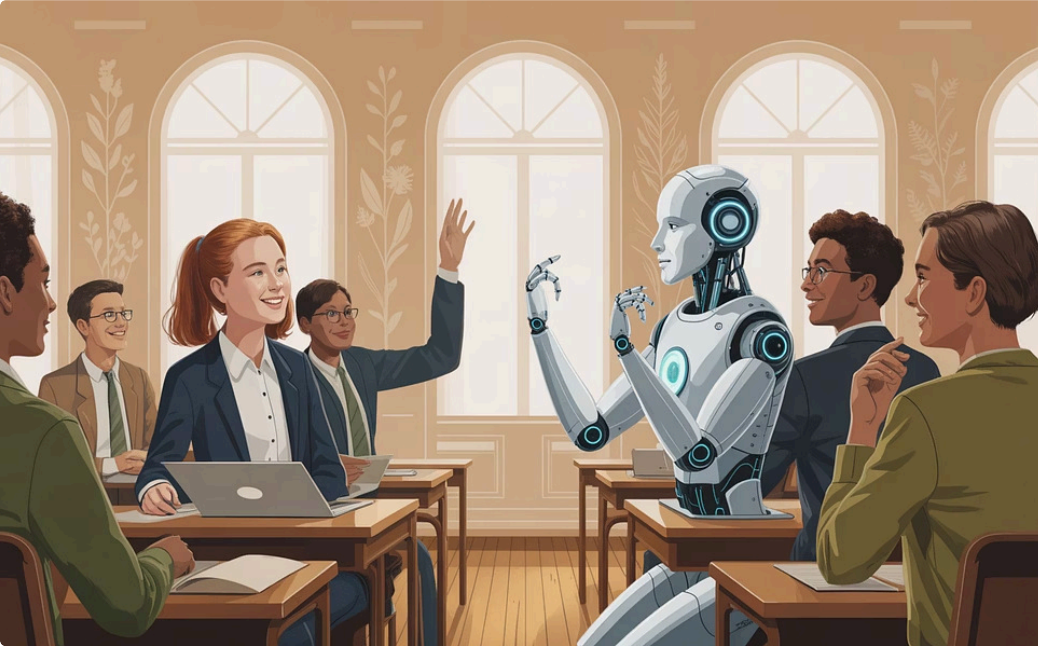
## 6. Testing and Validating

- Hyperparameter Tuning and Model Selection
- Data Mismatch



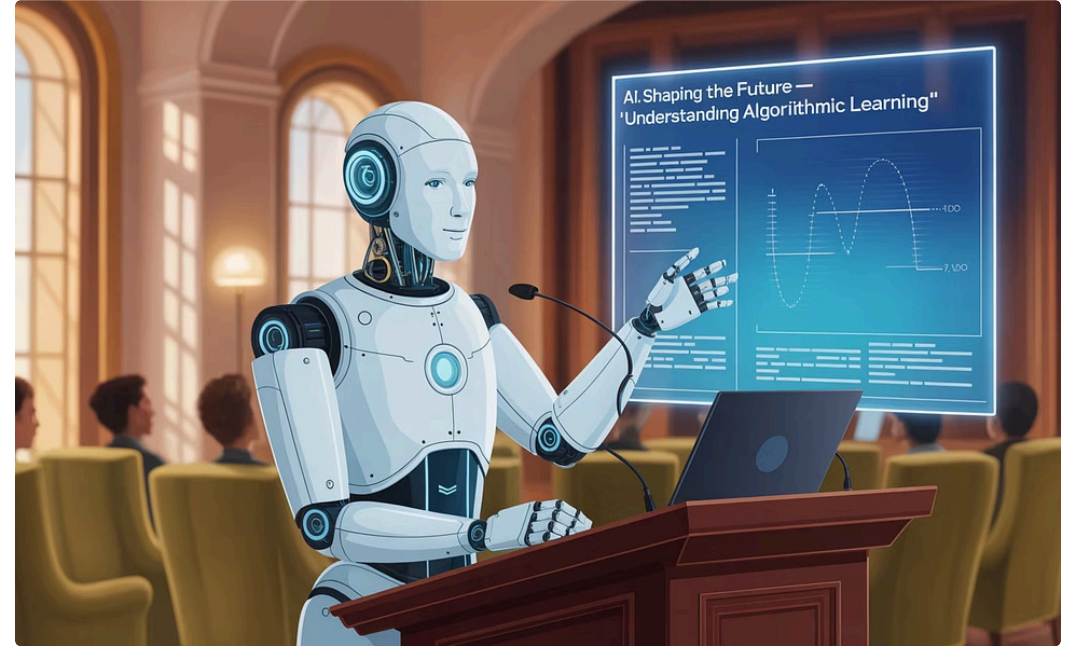
**Next: What Is Machine Learning?**

# Q&A



## Open Floor

Open floor for discussion or clarifications



## Discussion

Share your thoughts and questions about AI concepts